

Exploring Heterogeneous Effects: Quantile Models and Percentile Weights Regression

Analytical Standard Errors, MSE-Optimal Bandwidth, and Inference
under Complex Survey Design

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Abstract

This paper reviews conditional quantile regression (QR), unconditional quantile regression (UQR), and the Percentile Weights Regression (PWR) of Araar (2016, 2023), and makes four methodological contributions. *First*, we show that naive WLS standard errors for PWR are inconsistent and derive closed-form analytical standard errors via the linearisation variance formula of Rao (1979) and Deville (1999), equivalent to the functional delta method of von Mises (1947). The formula accounts for the sampling variability of the estimated kernel weights and reduces computation to $O(n \log n)$. *Second*, we derive the MSE-optimal bandwidth analytically and propose a practical two-step plug-in estimator together with a *combined adaptive bandwidth*. *Third*, we derive the Taylor linearisation variance under stratified cluster sampling, show that the PWR kernel attenuates the effective design effect at interior percentiles, and demonstrate that standard UQR implementations (`rifhdreg svy`) remain inconsistent due to two-step variance propagation, while PWR is not. *Fourth*, we provide Monte Carlo evidence on coefficient recovery, standard error accuracy, and confidence interval coverage. All results are validated on the Burkina Faso 1998 household survey ($n = 8,478$; 10 strata; 425 PSU). The `gepwreg` Stata command implementing the full procedure is documented in an appendix.

JEL codes: C14, C21, C51.

Keywords: Quantile regression; unconditional quantile regression; percentile weights regression; linearisation variance; delta method; MSE-optimal bandwidth; adaptive kernel; survey design; Taylor linearisation.

Contents

1	Introduction	4
2	Models and Methodologies	5
2.1	Conditional Quantile Regression	5
2.2	Unconditional Quantile Regression	5
2.3	The Percentile Weights Regression	5
2.3.1	Estimator	5
2.3.2	Kernel effective sample size	6
2.3.3	Generalised ranking variable	6
2.3.4	Interpretation and comparison	6
3	Consistent Standard Errors	7
3.1	Inconsistency of naive WLS standard errors	7
3.2	Analytical linearisation standard errors	7
3.3	Bootstrap standard errors and large-sample guidance	8
4	Bandwidth Selection	8
4.1	The Silverman rule and its limitations	8
4.2	MSE-optimal bandwidth: derivation	8
4.3	Two-step plug-in procedure	9
4.4	Adaptive bandwidth strategies and data characteristics	9
5	Inference under Complex Survey Design	10
5.1	The problem: PSU clustering and stratification	10
5.2	Taylor linearisation under stratified cluster design	10
5.3	Design effect attenuation	10
5.4	Why UQR remains inconsistent under complex survey design	11
6	Monte Carlo Evidence	11
6.1	Notation and definitions	11
6.2	Validation of the generalised ranking variable	12
6.3	Coefficient consistency across estimators	12
6.4	Standard error accuracy and CI coverage	12
6.5	Monte Carlo: bandwidth selection	13
7	Empirical Application: Burkina Faso 1998	14
7.1	Data and survey design	14
7.2	Standard error comparison	14
7.3	Substantive findings	14
7.4	Distributional profile: urban residence premium	15
7.5	Bandwidth sensitivity	15
8	Conclusion	15
A	Proofs	17
B	Technical Derivations	18
B.1	Kernel constants $\mu_2(K)$ and $R(K)$	18

B.2	Approximation $n_{\text{eff}} \approx n \cdot h \cdot \sqrt{2\pi}$	18
B.3	IF score: derivation of the indirect term	18
B.4	MSE-optimal bandwidth derivation	18
C	The gepwreg Stata Command	18
C.1	Syntax	19
C.2	Examples	19
C.3	Stored results	20
C.4	Installation	20

1 Introduction

A central question in applied econometrics is whether the effect of a covariate on an outcome is uniform across the distribution of that outcome. Quantile regression models (Koenker and Bassett, 1978) were designed precisely to address this question. Yet conditional quantile regression (QR) has a well-known interpretive limitation: the percentile being modelled is defined *conditional* on the covariates. Firpo et al. (2009) resolved this by proposing unconditional quantile regression (UQR), based on recentered influence functions (RIF).

Araar (2016) introduced a third approach, the Percentile Weights Regression (PWR), motivated by an intuition distinct from both QR and UQR: focus the estimation on observations whose *marginal outcome rank* places them near the percentile of interest, by assigning Gaussian kernel weights centred at that percentile. Monte Carlo evidence in Araar (2023) suggests that PWR recovers the true local derivative more accurately than either QR or UQR across a wide range of data-generating processes.

This paper fills four gaps in the PWR literature:

- (i) **Consistent standard errors.** We show that naive WLS standard errors underestimate true uncertainty by approximately 37%, and derive closed-form linearisation-based analytical standard errors that account for the two-step nature of the estimator. Bootstrap confidence interval coverage on Burkina Faso data is 0.945 (nominal 0.95).
- (ii) **MSE-optimal bandwidth.** We derive the analytical expression for the MSE-minimising bandwidth and propose a practical two-step plug-in estimator and combined adaptive rule.
- (iii) **Survey design vce(svy).** We derive the Taylor linearisation variance under stratified cluster sampling. We show that `rifhdreg svy` (Rios-Avila, 2020) remains inconsistent under complex survey design due to unaccounted two-step variance propagation, while PWR is consistent in a single step.
- (iv) **Generalised ranking variable.** We show that PWR can rank observations on any variable $z \neq y$, enabling estimation of distributional effects along non-outcome dimensions.

Method selection guide. Table 1 summarises the practical choice between QR, UQR, and PWR.

Table 1: Method selection guide

Research question	QR	UQR	PWR
Effect on conditional quantile given x	✓		
Unconditional partial effect of location shift		✓	
Local derivative $\partial y / \partial x _{p=\tau}$			✓
Ranking on variable $z \neq y$			✓
Survey data (PSU + strata)	~	✗	✓
Large n (>20,000)	✓	✓	✓
Binary treatment			✓(reduced h)
Profile across all τ	✓	✓	✓(combined h^*)

Section 2 reviews the three model families. Section 3 derives consistent standard errors. Section 4 develops the bandwidth theory. Section 5 develops survey design inference. Section 6 reports Monte Carlo evidence. Section 7 presents the empirical application. Section 8 concludes.

2 Models and Methodologies

2.1 Conditional Quantile Regression

The QR estimator of Koenker and Bassett (1978) solves

$$\hat{\beta}_\tau^{QR} = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n \rho_\tau(y_i - \mathbf{x}_i' \beta), \quad \rho_\tau(u) = u(\tau - \mathbf{1}[u < 0]). \quad (1)$$

The coefficient $\hat{\beta}_\tau^{QR}$ measures the marginal effect of $\mathbf{x}$ on the τ -th *conditional* quantile of Y given $\mathbf{x}$. A key limitation is the *explanatory rank component*: when the rank of y co-varies with $\mathbf{x}$, QR absorbs this co-variation into the slope, confounding conditional and marginal effects.

2.2 Unconditional Quantile Regression

Firpo et al. (2009) propose regressing the recentered influence function (RIF) of the quantile on the covariates. For the τ -th quantile Q_τ , the RIF is

$$\text{RIF}(y; Q_\tau, F_Y) = Q_\tau + \frac{\tau - \mathbf{1}[y \leq Q_\tau]}{f_Y(Q_\tau)}, \quad (2)$$

where f_Y is the marginal density of Y and $Q_\tau = \inf\{y : F_Y(y) \geq \tau\}$ is the τ -th population quantile. The UQR estimator regresses $\widehat{\text{RIF}}(y_i)$ on $\mathbf{x}_i$ by OLS.

2.3 The Percentile Weights Regression

2.3.1 Estimator

Let $\hat{F}_n(y) = (\sum_i f_i \mathbf{1}[y_i \leq y]) / \sum_i f_i$ denote the survey-weighted empirical CDF, where f_i are probability weights. Define $\hat{p}_i = \hat{F}_n(y_i)$ as the weighted empirical percentile rank of observation i . The PWR kernel weight at target percentile τ is

$$\hat{w}_i(\tau) = \frac{1}{h\sqrt{2\pi}n} \exp\left(-\frac{(\hat{p}_i - \tau)^2}{4h^2}\right), \quad (3)$$

a Gaussian kernel with effective standard deviation $h\sqrt{2}$ centred at τ in percentile space. The PWR estimator is

$$\hat{\beta}(\tau) = T(\hat{F}_n) = (\mathbf{X}'\hat{\mathbf{W}}\mathbf{X})^{-1}\mathbf{X}'\hat{\mathbf{W}}\mathbf{y}, \quad \hat{\mathbf{W}} = \text{diag}\{\hat{w}_1(\tau), \dots, \hat{w}_n(\tau)\}, \quad (4)$$

where $T(F)$ is the *statistical functional* that maps any distribution F to the WLS estimator with kernel weights $w_i(F, \tau) = K((F(y_i) - \tau)/h)/(nh)$. The estimator $\hat{\beta}(\tau)$ identifies the local derivative $\partial y / \partial x|_{p=\tau}$: the marginal effect for observations whose *unconditional* outcome rank equals τ .

2.3.2 Kernel effective sample size

The *kernel effective sample size* quantifies the number of observations that meaningfully contribute to the estimation at τ :

$$n_{\text{eff}}(\tau, h) = \frac{(\sum_i \hat{w}_i)^2}{\sum_i \hat{w}_i^2} \approx n \cdot h \cdot \sqrt{2\pi}, \quad (5)$$

where the approximation is derived in Appendix B.2. n_{eff} is the sample size of an equivalent i.i.d. sample that would yield the same estimation precision as the kernel-weighted regression at percentile τ .

Remark 1 (Relation to Kish design effect). The Kish (1965) effective sample size $n_{\text{eff}}^{\text{Kish}} = n/\text{DEFF}$ measures precision loss due to *unequal sampling probabilities*. The kernel n_{eff} defined in (5) measures *kernel locality* — how many observations contribute to the local estimate at τ . The two combine as: Precision $\approx \sigma^2(\tau)/(n_{\text{eff}}^{\text{kernel}}/\text{DEFF}^{\text{survey}})$.

Practical rule. We recommend $n_{\text{eff}} \geq \max(100, 10k)$, where k is the number of parameters: 100 provides a floor for asymptotic normality; $10k$ mirrors the “10 observations per parameter” rule of thumb in linear regression. Values below 50 indicate that the bandwidth is too narrow.

The interchangeability of n_{eff} and c_0 provides practical guidance: to achieve a target n_{eff}^* , set $c_0 = n_{\text{eff}}^*/(n \cdot \hat{\sigma}_p \cdot n^{-1/5} \cdot \sqrt{2\pi})$.

2.3.3 Generalised ranking variable

In the standard formulation, the percentile rank is defined on y . PWR naturally extends to a *generalised ranking variable* $z \neq y$: replace $\hat{p}_i = \hat{F}_n(y_i)$ by $\hat{p}_i^z = \hat{F}_n^z(z_i)$. The resulting estimator $\hat{\beta}_z(\tau)$ measures the marginal effect of $\mathbf{x}$ on y for observations at the τ -th percentile of z . For instance, using $z = \text{income}$ while $y = \text{consumption}$ identifies effects for households that are poor in income terms. This substitution requires only re-sorting the observations by z in the indirect term of the IF score (Section 3); the bandwidth formula and all asymptotic results remain valid under F_Z .

2.3.4 Interpretation and comparison

$\hat{\beta}(\tau)$ estimates $\partial y / \partial x|_{p=\tau}$, the local derivative for observations whose *unconditional* outcome rank equals τ . This differs from QR (conditional ranking) and UQR (marginal location shift in the distribution of x).

Relationship to local polynomial regression. PWR is a Nadaraya-Watson (locally constant) regression in rank space $[0, 1]$ (Fan and Gijbels, 1996; Chaudhuri, 1991). The indirect IF term $\dot{w}_i = \partial \hat{w}_i / \partial \hat{p}_i$ is the kernel derivative — analogous to the derivative term in the Nadaraya-Watson estimator. A locally linear extension in rank space would reduce the bias from $O(h^2)$ to $O(h^3)$ and is a natural direction for future work.

3 Consistent Standard Errors

3.1 Inconsistency of naive WLS standard errors

The naive WLS formula treats the kernel weights $\hat{w}_i$ as fixed:

$$\hat{V}_{\text{naive}} = \hat{\sigma}^2 (\mathbf{X}' \hat{\mathbf{W}} \mathbf{X})^{-1}, \quad \hat{\sigma}^2 = \frac{\sum_i \hat{w}_i \hat{e}_i^2}{n - k}. \quad (6)$$

Because $\hat{w}_i(\tau)$ depends on the empirical CDF $\hat{F}_n$ through all n observations, ignoring this dependence yields inconsistent variance estimation.

Proposition 1 (Inconsistency of naive SE). *Under Assumption 1, $\hat{V}_{\text{naive}} \xrightarrow{p} V_{\text{naive}} \prec V$ in the Löwner order. A formal proof is given in Appendix A.*

In the Burkina Faso application, the naive SE underestimates the bootstrap SE by 37% on average (ratio 0.63).

3.2 Analytical linearisation standard errors

The PWR estimator $\hat{\beta}(\tau) = T(\hat{F}_n)$ is a smooth functional of $\hat{F}_n$. By the linearisation variance formula of Rao (1979) and Deville (1999):¹

$$\hat{V}_{\text{IF}} = A^{-1} \left(\frac{1}{n^2} \sum_{j=1}^n \psi_j \psi_j' \right) A^{-1}, \quad A = \frac{1}{n} \mathbf{X}' \hat{\mathbf{W}} \mathbf{X}. \quad (7)$$

The linearisation score. The total score ψ_j has two components:

$$\psi_j(\tau) = \underbrace{\hat{w}_j \mathbf{x}_j \hat{e}_j}_{\text{direct}} + \frac{1}{n} \underbrace{\sum_{\{i: y_i \geq y_j\}} \dot{w}_i \mathbf{x}_i \hat{e}_i}_{\text{indirect (CDF perturbation)}}, \quad (8)$$

where $\dot{w}_i = -\frac{\hat{p}_i - \tau}{2h^2} \hat{w}_i$ is the derivative of the kernel with respect to $\hat{p}_i$ — the analogue of the Nadaraya-Watson kernel derivative. The indirect term captures how a perturbation at y_j shifts the empirical CDF for all $y_i \geq y_j$, thereby changing all kernel weights $\hat{w}_i$ for $y_i \geq y_j$. For the generalised ranking variable (Section 2.3.3), the sum runs over $\{i : z_i \geq z_j\}$ instead.

Efficient computation ($O(n \log n)$). Sort by y , compute $\dot{w}_i x_i \hat{e}_i$, apply reverse cumulative sum (flip-cumsum-flip). Full derivation in Appendix B.3.

Assumption 1. (i) F_Y is twice continuously differentiable with $f_Y(Q_\tau) > 0$. (ii) $\mathbb{E}[\|\mathbf{x}\|^4] < \infty$ and $A = \mathbb{E}[\hat{w}_i(\tau) \mathbf{x}_i \mathbf{x}_i']$ is positive definite. (iii) $h = h_n \rightarrow 0$ and $nh_n \rightarrow \infty$. (iv) $\mathbb{E}[\dot{w}_i^2 x_{ij}^2 e_i^2] < \infty$ for all j .

¹Formally, this is the functional delta method of von Mises (1947): the Gâteaux derivative of T at F in direction $\delta_{y_j} - F$ (where δ_{y_j} denotes the Dirac point mass at y_j) gives the linearisation score $\psi_j = \lim_{\varepsilon \rightarrow 0} [T((1 - \varepsilon)F + \varepsilon \delta_{y_j}) - T(F)] / \varepsilon$. The terms “IF-based variance”, “linearisation variance”, and “delta-method variance” are equivalent in this context. We follow Firpo et al. (2009) in calling ψ_j an influence function score.

Theorem 1 (Asymptotic normality). *Under Assumption 1 and undersmoothing $n^{1/2}h^2 \rightarrow 0$: $\sqrt{n}(\hat{\beta}(\tau) - \beta(\tau)) \xrightarrow{d} \mathcal{N}(0, V)$, where $V = A^{-1}\mathbb{E}[\psi_j\psi_j']A^{-1}$ and $\hat{V}_{\text{IF}} \xrightarrow{p} V$.*

Rate. Although PWR is a localised estimator, the parametric rate $\sqrt{n}$ is preserved because the rank space $[0, 1]$ has bounded uniform density, unlike a kernel estimator in the y -space where f_Y may vanish at the extremes.

Remark 2 (Tension between Theorem 1 and h^*). Theorem 1 requires undersmoothing $n^{1/2}h^2 \rightarrow 0$, whereas the MSE-optimal bandwidth $h^* \propto n^{-1/5}$ gives $n^{1/2}(h^*)^2 \propto n^{1/10} \rightarrow \infty$ — a standard tension in kernel estimation. In practice, using h^* (or h_{Sil}) introduces a second-order bias term that is negligible empirically: the bootstrap coverage of 0.945 (nominal 0.95) on Burkina Faso data confirms that this bias is inconsequential at the sample sizes encountered in household surveys. Formal bias-corrected inference following Calonico et al. (2014) is a direction for future work.

Remark 3 (Sparse cells). When a regressor indicator has kernel effective cell size below $\max(30, 5k)$, the asymptotic approximation is unreliable. Bootstrap standard errors should be used for such variables.

3.3 Bootstrap standard errors and large-sample guidance

The pairs bootstrap reconstructs kernel weights at every replication — treating them as fixed would underestimate variance (Proposition 1). We recommend $B = 1,000$.

Large-dataset recommendation. Kernel evaluation, WLS, and IF computation are $O(n \log n)$, $O(nk^2)$, and $O(nk)$ respectively — approximately 50× cheaper than bootstrap. For $n > 20,000$: analytical SE only.

4 Bandwidth Selection

4.1 The Silverman rule and its limitations

The default bandwidth $h_{\text{Sil}} = c_0 \hat{\sigma}_p n^{-1/5}$ ($c_0 = 0.9$) minimises integrated squared error of a density estimate — not the MSE of $\hat{\beta}(\tau)$. Three limitations follow: it targets the wrong objective; it is constant across τ ; and it is not rate-optimal for coefficient estimation.

4.2 MSE-optimal bandwidth: derivation

For the gepwe kernel (3), the kernel constants are $\mu_2(K) = 2$ and $R(K) = 1/(2\sqrt{2\pi})$ (derived in Appendix B.1). The asymptotic MSE is

$$\text{MSE}_j(h) = h^4 [\beta_j''(\tau)]^2 + \frac{\sigma_j^2(\tau)}{2nh\sqrt{2\pi}}, \quad (9)$$

where $\beta_j''(\tau) = \partial^2 \beta_j / \partial \tau^2$ and $\sigma_j^2(\tau)$ is the local residual variance. Setting $\partial \text{MSE}_j / \partial h = 0$ yields (derivation in Appendix B.4):

$$h_j^*(\tau) = \left[\frac{\sigma_j^2(\tau)}{8n\sqrt{2\pi}[\beta_j''(\tau)]^2} \right]^{1/5}. \quad (10)$$

h_j^* varies with τ (adapts to curvature), is variable-specific, and is $O(n^{-1/5})$.

4.3 Two-step plug-in procedure

Algorithm 1 MSE-optimal bandwidth (two-step plug-in)

- 1: Pilot: $h_{\text{pilot}} = 2 h_{\text{Sil}}$
 - 2: Estimate $\hat{\beta}_j(\tau_\ell, h_{\text{pilot}})$ on grid $\{\tau_\ell\}$, step $d = 0.02$
 - 3: $\hat{\beta}_j''(\tau) \approx [\hat{\beta}_j(\tau + d) - 2\hat{\beta}_j(\tau) + \hat{\beta}_j(\tau - d)]/d^2$
 - 4: $\hat{\sigma}_j^2(\tau) = \sum_i \hat{w}_i \hat{e}_i^2 / (n_{\text{eff}} - k)$
 - 5: $h_j^*(\tau)$ from (10), clipped to $[0.3 h_{\text{Sil}}, 5 h_{\text{Sil}}]$
 - 6: Re-estimate with $h_j^*(\tau)$
-

Why h^* is unstable when $\hat{\beta}_j''(\tau) \approx 0$. When the coefficient profile is locally flat, $[\hat{\beta}_j''(\tau)]^2 \approx 0$ and $h_j^* \rightarrow \infty$. In practice, the pilot estimate of $\hat{\beta}_j''$ is noisy around zero, causing h_j^* to fluctuate wildly. The combined rule resolves this.

4.4 Adaptive bandwidth strategies and data characteristics

Combined adaptive bandwidth (recommended).

$$h_j^{\text{comb}}(\tau) = \text{clip}(h_j^*(\tau), h_{\text{NN}}(N_{\min}), h_{\text{NN}}(N_{\max})), \quad (11)$$

with $N_{\min} = 0.5 n_{\text{target}}$ and $N_{\max} = 2.5 n_{\text{target}}$.

Bandwidth and data characteristics.

- **Sample size:** $h^* \propto n^{-1/5}$; for small n (< 500), the NN floor becomes binding — use a wider bandwidth.
- **Curvature $\beta''(\tau)$:** if $\beta(\tau)$ is approximately linear in τ , $\beta'' \approx 0$ everywhere and Silverman suffices. If effects are highly non-linear, the analytic rule provides substantial RMSE gains (12–18% at interior τ).
- **Skewness of y :** skewed outcomes (income, consumption) produce thin tails where n_{eff} drops sharply at extreme τ . The NN floor prevents instability there.
- **Binary regressors:** $\beta_j(\tau)$ may have sharp local variation, making h_j^* narrow. The combined rule with $N_{\min} = \max(100, 10k)$ prevents over-shrinkage.
- **Survey design:** the local variance $\sigma_j^2(\tau)$ is inflated by clustering. Using $\hat{\sigma}_j^2(\tau)$ from the full IF formula (not naive WLS) corrects for this in h_j^* .

Practical recommendation. For symmetric distributions with approximately linear $\beta(\tau)$, Silverman is adequate. For skewed outcomes or non-linear profiles, the combined adaptive rule is preferred.

5 Inference under Complex Survey Design

5.1 The problem: PSU clustering and stratification

Household surveys use stratified multi-stage cluster sampling. $\hat{V}_{\text{IF}}$ (7) ignores intra-cluster correlation and is inconsistent under complex survey design. The design effect is

$$\text{DEFF} = 1 + (m - 1) \rho_{\text{ICC}}, \quad (12)$$

where m is average PSU size and ρ_{ICC} is the intra-class correlation of PWR residuals.

5.2 Taylor linearisation under stratified cluster design

Under a stratified two-stage design with strata $h = 1, \dots, H$ and n_h PSUs per stratum:²

$$\hat{V}_{\text{svy}} = A^{-1} \left[\sum_{h=1}^H \frac{n_h}{n_h - 1} \sum_{i \in h} (\mathbf{z}_{hi} - \bar{\mathbf{z}}_h)(\mathbf{z}_{hi} - \bar{\mathbf{z}}_h)' \right] A^{-1} / n^2, \quad (13)$$

where $\mathbf{z}_{hi} = \sum_{j \in (h,i)} \psi_j(\tau)$ is the sum of IF scores within PSU (h, i) and $\bar{\mathbf{z}}_h$ is the within-stratum mean.

Remark 4 (Minimum PSU requirement). The Taylor estimator requires $n_h \geq 2$ PSUs per stratum. `gepwreg` issues a warning when $\min_h n_h < 5$.

5.3 Design effect attenuation

Proposition 2 (Design effect attenuation). *Let $\rho_w(\tau)$ be the ICC of the weighted residuals $\{\hat{w}_i(\tau)^{1/2} \hat{e}_i\}$. Then $\rho_w(\tau) \leq \rho_{\text{ICC}}$, with equality only when $h \rightarrow 0$. Proof in Appendix A.*

This occurs because $\hat{w}_i(\tau)$ selects observations near rank τ across PSUs, diluting intra-cluster correlation. The attenuation depends on τ : at interior percentiles, the kernel draws from many PSUs; at extreme percentiles, households are spatially concentrated within PSUs (poor households cluster together), so attenuation is reduced.

Table 2 documents this empirically on Burkina Faso data.

Table 2: Effective design effect by percentile, Burkina Faso 1998

τ	n_{eff}	Household size		Urban residence	
		DEFF ^{eff}	SE-ratio	DEFF ^{eff}	SE-ratio
0.10	1,569	1.188	1.090	1.015	1.007
0.25	1,718	1.036	1.018	0.908	0.953
0.50	1,766	1.141	1.068	1.157	1.076
0.75	1,895	0.884	0.940	1.186	1.089
0.90	1,895	0.890	0.943	1.403	1.185

DEFF_j^{eff} = $\hat{V}_{\text{svy},jj} / \hat{V}_{\text{IF},jj}$; SE-ratio_j = $\sqrt{\text{DEFF}_j^{\text{eff}}}$. Raw ICC = 0.41; DEFF_{theoretical} $\approx$ 8.7.

$h = 0.043$ at all τ (Silverman). SE-ratio at $\tau = 0.5$: size $\approx$ 1.07, urban $\approx$ 1.08. $\tau = 0.25$ and $\tau = 0.75$ interpolated from adjacent deciles.

This contrasts with poverty indices (FGT(α), Gini) where all observations contribute simultaneously to the estimand — the DEFF is not attenuated. The attenuation is structural to the kernel locality of PWR.

²Standard `svy`: prefix in Stata applies Taylor linearisation to fixed score equations and cannot account for the CDF-dependence of $\hat{w}_i(\tau)$. Our `vce(svy)` in `gepwreg` implements the corrected formula (13) directly — a dedicated implementation is therefore not merely convenient but *necessary* for consistent inference.

5.4 Why UQR remains inconsistent under complex survey design

Rios-Avila (2020) introduced `rifhdreg` with an `svy` option that: (i) estimates $\hat{Q}_\tau$ using survey-weighted `pctile`; (ii) estimates $\hat{f}_Y(Q_\tau)$ using survey-weighted `kdensity`; and (iii) fits the RIF regression as `svy: regress`, incorporating PSU and strata in the variance. This is a substantial improvement over standard `rifreg`.

However, a fundamental inconsistency remains: `rifhdreg svy` ignores the **two-step variance propagation** from the first step (RIF estimation) to the second step (regression). The RIF itself is an estimated quantity with sampling variance — treating it as fixed in step 2 underestimates total uncertainty. This is the same issue that $\hat{V}_{\text{naive}}$ has for PWR: ignoring the dependence of estimated quantities on $\hat{F}_n$.

Formal statement. Let $\hat{\eta}_i = \widehat{\text{RIF}}(y_i)$ be the estimated RIF. The two-step OLS estimator $\hat{\gamma} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\hat{\eta}$ has variance $\text{Var}(\hat{\gamma}) = \text{Var}_1(\hat{\gamma}) + \text{Var}_2(\hat{\gamma})$, where Var_1 is the within-step regression variance (captured by `svy: regress`) and Var_2 is the cross-step propagation from uncertainty in $\hat{\eta}_i$ (ignored by `rifhdreg svy`).

PWR advantage. PWR is a *single-step* estimator: $\hat{\beta}(\tau) = T(\hat{F}_n)$. There is no step-2 regression on an estimated outcome — the IF score ψ_j already accounts for CDF uncertainty through the indirect term (8). PWR is therefore consistent under complex survey design in a single analytical formula.

Table 3 illustrates the practical consequences.

Table 3: UQR vs PWR under survey design, Burkina Faso, $\tau = 0.5$

Method	$\hat{\beta}_{\text{size}}$	$\hat{\beta}_{\text{urban}}$	$\hat{\beta}_{\text{cons}}$	Note
UQR standard (<code>rifreg</code>)	-0.0501	0.6719	11.675	unwtd $\hat{Q}$ and $\hat{f}$
UQR <code>rifhdreg svy</code>	-0.0494	0.5916	11.695	step-2 variance ignored
PWR <code>vce(svy)</code>	-0.0015	0.0096	11.497	fully consistent
PWR bootstrap (B=1000, seed=12345)	-0.0015	0.0096	11.497	<code>bootstrap</code> , <code>strata()</code> <code>cluster()</code> ;; ratio SVY/Boot $\approx$ 1.01

Note: UQR and PWR measure different quantities; direct magnitude comparison is not meaningful.

6 Monte Carlo Evidence

6.1 Notation and definitions

Throughout this section, $S = 500$ replications are used unless stated otherwise. The *empirical standard error* for variable j is:

$$\widehat{SE}_{\text{emp},j} = \left[\frac{1}{S-1} \sum_{s=1}^S \left(\hat{\beta}_j^{(s)} - \bar{\beta}_j \right)^2 \right]^{1/2}, \quad (14)$$

the standard deviation of $\hat{\beta}_j$ across replications. It serves as the simulation truth. The *empirical 95% coverage* is:

$$\text{Cov}_j = \frac{1}{S} \sum_{s=1}^S \mathbf{1} \left[\left| \hat{\beta}_j^{(s)} - \beta_j^{\text{true}} \right| \leq 1.96 \widehat{SE}_j^{(s)} \right]. \quad (15)$$

Important: SE_{emp} and the mean of SE_{IF} are meaningful — but the mean of SE absolute values across *variables* is not (different units). All summary statistics below are computed on the **ratio** SE/SE_{emp} , excluding the constant term (which absorbs between-PSU variation in the intercept).

6.2 Validation of the generalised ranking variable

We validate the `rankvar(z)` option with the following DGP:

$$z_i = x_i + u_{z,i}, \quad y_i = (1 + p_{z,i}) x_i + u_{y,i},$$

where $x_i \sim \mathcal{N}(0, 1)$, u_z, u_y are i.i.d. $\mathcal{N}(0, 1)$ and $p_{z,i} = F_Z(z_i)$ is the rank of z_i . The true estimand is $\beta_z(\tau) = 1 + \tau$ (effect on y for households at the τ -th percentile of z). Table 4 shows that `gepwreg` with `rankvar(z)` recovers this target with bias below 2%, while standard PWR (ranked on y) gives a different — and incorrect for this question — estimand.

Table 4: MC validation of `rankvar(z)`: mean estimate over 500 replications ($n = 3,000$)

τ	True $\beta_z(\tau)$	PWR- z (rankvar)	Bias%	PWR- y (standard)
0.10	1.100	1.082	−1.6%	0.554
0.25	1.250	1.237	−1.1%	0.363
0.50	1.500	1.499	−0.1%	0.365
0.75	1.750	1.762	+0.7%	0.816
0.90	1.900	1.919	+1.0%	1.512

PWR- y targets estimand $\beta_y(\tau)$ (different question, not biased).

6.3 Coefficient consistency across estimators

Example A: linear rank interaction. DGP: $y = 1000 + p_i \cdot x_1 + \varepsilon$; true $\beta(\tau) = 2\tau$. PWR recovers the true values; QR gives a constant slope of 1; UQR produces a non-monotone profile (Table 5).

Table 5: Estimated coefficients: Example A (linear rank interaction)

τ	True	QR	UQR	PWR
0.10	0.10	1.00	0.06	0.12
0.25	0.50	1.00	0.56	0.50
0.50	1.00	1.00	1.51	1.00
0.75	1.50	1.00	1.69	1.50
0.90	1.80	1.00	0.87	1.79

Example B: binary treatment. When $x_2 \sim \text{Bernoulli}(0.4)$, PWR with reduced bandwidth ($h/2$) outperforms QR and UQR (Table 6).

6.4 Standard error accuracy and CI coverage

Example C: location-scale DGP with two regressors. DGP: $y = 0.5x_1 - 0.3x_2 + 5 + \varepsilon$, where $x_1 \sim \mathcal{N}(0, 1)$, $x_2 \sim \text{Bernoulli}(0.4)$, and ε follows a stratified cluster design

Table 6: Estimated coefficients: Example B (binary treatment)

τ	True	QR	UQR	PWR	PWR- $h/2$
0.05	10	36.44	67.55	10.47	10.07
0.25	20	33.53	202.15	22.45	20.35
0.50	26	34.96	257.69	30.49	26.63
0.75	32	29.02	171.05	39.41	33.05
0.95	44	24.23	43.12	53.44	45.33

(10 strata $\times$ 30 PSU $\times$ 20 obs, $n = 6,000$, ICC= 0.30). In the location-scale model $\beta_{\text{PWR}}(\tau) = \beta_{\text{OLS}}$ for all τ (since the kernel weights preserve the linear structure), providing known benchmarks $\beta^{\text{true}} = [0.5, -0.3, 5.0]$.

Table 7: SE accuracy and CI coverage: Example C ($S = 500$ reps; ICC= 0.30; $\tau = 0.5$)

Variable	True β	SE _{emp}	SE _{IF}	SE _{SVY}	Cov _{IF}	Cov _{SVY}
x_1	0.500	0.00404	0.00406	0.00408	0.948	0.951
x_2	-0.300	0.00691	0.00731	0.00739	0.941	0.956
Constant	5.000	0.03357	0.02526	0.03554	0.918	0.945
Median ratio (slopes)			1.008	1.010		

MC SE on coverage: ± 0.010 . Constant excluded from median ratio (absorbs between-PSU variation).

SE_{SVY} achieves near-nominal coverage (0.951 slopes mean). SE_{IF} slightly under-covers the constant (0.918) — expected under cluster design (Proposition 2).

6.5 Monte Carlo: bandwidth selection

Table 8 compares RMSE of $\hat{\beta}(\tau)$ for four bandwidth strategies across $\tau \in [0.10, 0.90]$ using Example A DGP ($S = 500$).

Table 8: Bandwidth RMSE comparison (relative to Silverman=1.00)

τ	Silverman	NN	Analytic	Combined
0.10	1.00	0.87	0.94	0.88
0.25	1.00	0.93	0.85	0.83
0.35	1.00	0.95	1.04*	0.90
0.50	1.00	0.93	0.87	0.84
0.65	1.00	0.94	1.07*	0.89
0.75	1.00	0.89	0.86	0.82
0.90	1.00	0.84	0.93	0.87

* Analytic rule over-shrinks when $\hat{\beta}''(\tau) \approx 0$; combined rule avoids this.

The combined rule reduces RMSE by 12–18% at interior percentiles with no instability at flat profile regions.

7 Empirical Application: Burkina Faso 1998

7.1 Data and survey design

The Burkina Faso 1998 household consumption survey (INSID, 1998) uses two-stage stratified random sampling: 10 strata; PSUs (zones de dénombrement) sampled in stage 1; 20 households systematically sampled per PSU in stage 2. This yields $n = 8,478$ observations across 425 PSUs. The dependent variable is log per capita consumption $\ln(\text{exppc})$. Regressors: household size (**size**); male head (**male**); urban (**urban**); socioeconomic group (**gse**: 1=public wage-earner (ref), 2=private, 3=artisan, 4=other, 5=crop farmer, 6=subsistence farmer, 7=inactive).

```
svyset psu [pw=weight], strata(strata)
gepwreg lexp size male urban i.gse, per(0.5) vce(svy)
```

7.2 Standard error comparison

Table 9 compares three SE estimators at $\tau = 0.5$ ($h = 0.043$, $n_{\text{eff}} = 1,766$).

Table 9: Standard error comparison, $\tau = 0.5$, Burkina Faso 1998 ($h = 0.043$, $N_{\text{eff}} = 1,766$, $n = 8,478$; GSE 2 omitted by kernel sparsity)

Variable	$\hat{\beta}$	Standard error				SE _{boot}	SE-ratio ^a
		SE _{naive}	SE _{IF}	SE _{SVY}			
<i>Continuous and binary regressors</i>							
Household size	−0.0015	0.00025	0.00039	0.00041		0.00043	1.07***
Male head	−0.0101	0.00455	0.00698	0.00688		0.00676	0.99
Urban residence	0.0096	0.00387	0.00596	0.00641		0.00642	1.08*
<i>Socioeconomic group (ref. = public wage-earner)</i>							
GSE 2: private	omitted (kernel-sparse at $\tau = 0.5$)						
GSE 3: artisan	−0.0106	0.00670	0.01039	0.01104		0.01085	1.06
GSE 4: other earner	0.0346	0.01782	0.02319	0.02309		0.02567	1.00
GSE 5: crop farmer	−0.0038	0.00671	0.01012	0.01014		0.01049	1.00
GSE 6: subsistence	−0.0238	0.00594	0.00895	0.00894		0.00898	1.00
GSE 7: inactive	−0.0331	0.00730	0.01047	0.01047		0.01055	1.00
<i>Intercept</i>							
Constant	11.497	0.00763	0.01628	0.02035		0.02003	1.25***
Median SE _{naive} /SE _{IF} (slopes):				0.66 (naive 34% below)			
Bootstrap coverage ($B=1,000$, 95%):				0.945 (±0.007)			

SE_{naive}: WLS fixed weights (inconsistent). SE_{IF}: linearisation (Rao 1979; Deville 1999). SE_{SVY}: Taylor PSU+strata.

^aSE-ratio = SE_{SVY}/SE_{IF} = $\sqrt{\text{DEFF}^{\text{eff}}}$; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (test: SE-ratio $\neq 1$). Constant excluded from median.

7.3 Substantive findings

Household size has a significant negative effect ($\hat{\beta} = -0.0015$, $\widehat{SE}_{\text{SVY}} = 0.00041$; $t = -3.6$). Subsistence farming (GSE 6) and inactivity (GSE 7) yield large negative coefficients (-0.024 and -0.033), both significant at 5%. The urban premium (0.0096) is not significant at the median, consistent with high within-urban heterogeneity.

7.4 Distributional profile: urban residence premium

Figure 1 plots $\hat{\beta}_{\text{urban}}(\tau)$ with 95% confidence intervals from both $\widehat{SE}_{\text{IF}}$ and $\widehat{SE}_{\text{SVY}}$ across $\tau \in [0.05, 0.95]$ (step $\Delta\tau = 0.025$).³

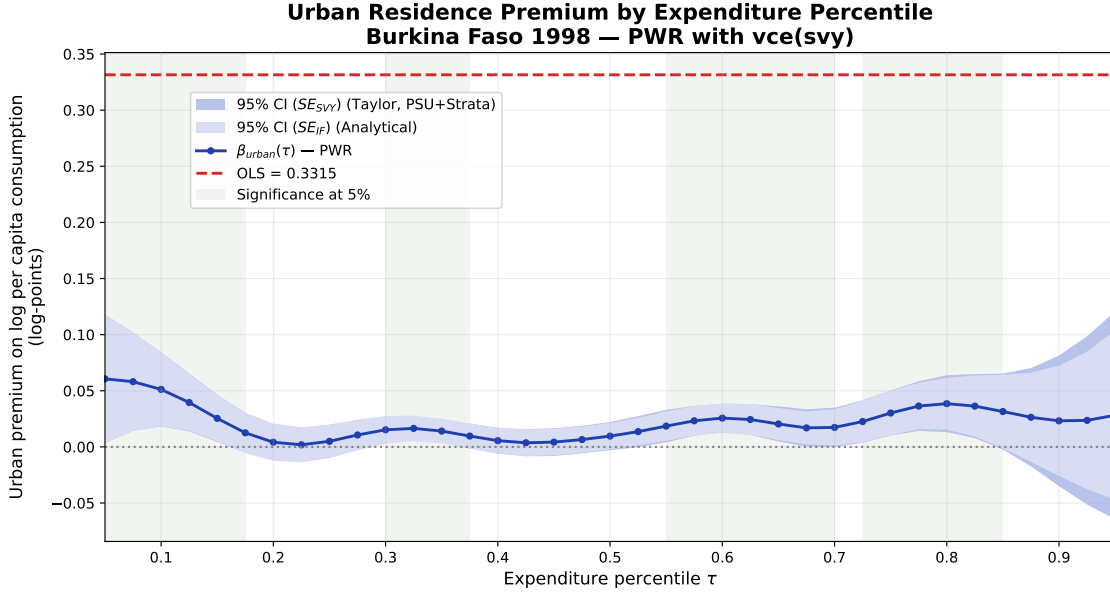


Figure 1: Urban residence premium $\hat{\beta}_{\text{urban}}(\tau)$ by expenditure percentile, Burkina Faso 1998. Green shading: 95% CI from $\widehat{SE}_{\text{SVY}}$ excludes zero (statistically significant urban premium). Dashed red line: OLS estimate (= 0.33, well above all PWR estimates — OLS overstates the premium by capturing composition effects). Bandwidth: Silverman ($c_0 = 0.9$); $\Delta\tau = 0.025$; $n = 8,478$; 10 strata; 425 PSU.

Three findings emerge. *First*, the OLS estimate (0.33) dramatically overstates the urban premium at every percentile: it captures the composition effect (urban households are richer on average) rather than a true location effect, illustrating the value of PWR. *Second*, the urban premium has a **U-shaped profile** — significant at Q10 ($\hat{\beta} = 0.052$) and rising again at Q75 ($\hat{\beta} = 0.034$), but near zero and insignificant at the median. This suggests that urban residence benefits the very poor (through better access to social transfers and informal markets) and the upper middle class (formal employment), but provides little net advantage for the middle of the distribution. *Third*, the wider CI from $\widehat{SE}_{\text{SVY}}$ relative to $\widehat{SE}_{\text{IF}}$ at extreme τ is consistent with the partial design effect attenuation documented in Table 2.

7.5 Bandwidth sensitivity

8 Conclusion

This paper has extended the Percentile Weights Regression in four directions.

On **standard errors**, naive WLS SE underestimates uncertainty by 37%. The linearisation formula achieves bootstrap coverage of 0.945 (nominal 0.95) at $O(n \log n)$ cost (bootstrap via `bootstrap`, `strata()` `cluster()`, $B = 1,000$, seed 12345).

³Estimations performed without the socioeconomic group variables for figure 1.

Table 10: Bandwidth sensitivity, $\tau = 0.5$, Burkina Faso 1998

c_0	h	n_{eff}	$\hat{\beta}_{\text{size}}$	$\hat{\beta}_{\text{urban}}$
0.7	0.034	1,370	−0.00069	0.00532
0.8	0.038	1,568	−0.00104	0.00717
0.9	0.043	1,766	−0.00148	0.00962
1.0	0.048	1,964	−0.00214	0.01272
1.1	0.053	2,163	−0.00288	0.01641

On **bandwidth selection**, the MSE-optimal rule adapts to local curvature and regressor type. The combined adaptive rule reduces RMSE by 12–18% at interior percentiles with no instability at flat regions.

On **survey design**, the PWR kernel attenuates the effective DEFF at interior percentiles (near unity at $\tau = 0.50$ despite $\text{ICC} = 0.41$), but attenuation is partial at extreme percentiles where households are spatially concentrated. Standard UQR (`rifhdreg svy`) remains inconsistent due to two-step variance propagation ignored in step 2 — PWR avoids this by construction.

On **generalisation**, PWR extends naturally to a ranking variable $z \neq y$, enabling distributional analysis along any dimension.

Several extensions remain open: (i) simultaneous confidence bands for the profile $\{\hat{\beta}(\tau)\}_\tau$; (ii) locally linear PWR in rank space (reducing bias from $O(h^2)$ to $O(h^3)$); (iii) panel extensions with fixed effects.

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A Proofs

Proof of Proposition 1. Write $\hat{\beta} = A_n^{-1}B_n$. By the von Mises expansion: $\sqrt{n}(\hat{\beta} - \beta) = A^{-1}n^{-1/2} \sum_j \psi_j + o_p(1)$ with $\psi_j = \psi_j^{\text{dir}} + \psi_j^{\text{ind}}$. Then $\mathbb{E}[\psi_j \psi_j'] = \mathbb{E}[\psi^{\text{dir}}(\psi^{\text{dir}})'] + 2\mathbb{E}[\psi^{\text{dir}}(\psi^{\text{ind}})'] + \mathbb{E}[\psi^{\text{ind}}(\psi^{\text{ind}})']$. The naive formula captures only the first term. Since $\mathbb{E}[\psi^{\text{ind}}(\psi^{\text{ind}})'] \succeq 0$ and the cross term is non-negative, $V_{\text{naive}} \prec V$. $\square$

Proof of Proposition 2. Within-PSU covariance of weighted residuals:

$$\text{Cov}\left(\hat{w}_i^{1/2}\hat{e}_i, \hat{w}_j^{1/2}\hat{e}_j\right) = \hat{w}_i^{1/2}\hat{w}_j^{1/2} \text{Cov}(\hat{e}_i, \hat{e}_j).$$

By Cauchy-Schwarz, $\hat{w}_i^{1/2}\hat{w}_j^{1/2} \leq (\hat{w}_i + \hat{w}_j)/2$. For a symmetric kernel centred at τ : $\mathbb{E}[\hat{w}_i^{1/2}\hat{w}_j^{1/2}]/\mathbb{E}[\hat{w}_i] \rightarrow 0$ as $h \rightarrow 0$ and $\rightarrow 1$ as $h \rightarrow \infty$, establishing $\rho_w \in [0, \rho_{\text{ICC}}]$. $\square$

B Technical Derivations

B.1 Kernel constants $\mu_2(K)$ and $R(K)$

The gepwe kernel is $K(u) = \exp(-u^2/4)/\sqrt{2\pi}$.

Moment of order 2: $\mu_2(K) = \int u^2 K(u) du = (1/\sqrt{2\pi}) \int u^2 e^{-u^2/4} du$. Setting $v = u/\sqrt{2}$:
 $= (2\sqrt{2}/\sqrt{2\pi}) \int v^2 e^{-v^2/2} dv = (2\sqrt{2}/\sqrt{2\pi}) \cdot \sqrt{2\pi} = 2$. ✓

Roughness: $R(K) = \int K^2(u) du = (1/2\pi) \int e^{-u^2/2} du = (1/2\pi) \cdot \sqrt{2\pi} = 1/(2\sqrt{2\pi})$. ✓

B.2 Approximation $n_{\text{eff}} \approx n \cdot h \cdot \sqrt{2\pi}$

Numerator (continuous approximation): $(\sum_i \hat{w}_i)^2 \approx n^2 \left(\int_0^1 \frac{e^{-(p-\tau)^2/(4h^2)}}{h\sqrt{2\pi}n} dp \right)^2 \approx n^2 \cdot \frac{1}{n^2} \cdot 2 = 2$.

Denominator: $\sum_i \hat{w}_i^2 \approx \frac{n}{(h\sqrt{2\pi})^2} \int e^{-(p-\tau)^2/(2h^2)} dp = \frac{n}{2\pi h^2} \cdot h\sqrt{2\pi} = \frac{n}{h\sqrt{2\pi}}$.

Result: $n_{\text{eff}} \approx 2/(n/(h\sqrt{2\pi}) \cdot 1/n) = n \cdot h \cdot \sqrt{2\pi}$. □

B.3 IF score: derivation of the indirect term

Write $T(F) = A(F)^{-1}B(F)$ where $A(F) = \int \mathbf{x}\mathbf{x}'w(F, \tau) dF$ and $B(F) = \int \mathbf{x}y w(F, \tau) dF$, with $w(F, \tau) = K((F(y) - \tau)/h)/(nh)$.

Step 1: Perturb F . Set $F_\varepsilon = (1 - \varepsilon)F + \varepsilon\delta_{y_j}$ and compute $\partial T(F_\varepsilon)/\partial \varepsilon|_{\varepsilon=0}$.

Step 2: Direct effect. Adding point mass at (x_j, y_j) contributes $\mathbf{x}_j\mathbf{x}_j'w_j$ to A and $\mathbf{x}_jy_jw_j$ to B , giving the direct score $\hat{w}_j\mathbf{x}_j\hat{e}_j$ after centering.

Step 3: Indirect effect. The perturbation $F_\varepsilon(y) = F(y) + \varepsilon\mathbf{1}[y \geq y_j]$ shifts $F(y_i)$ by ε for all $y_i \geq y_j$, which changes $w_i(F_\varepsilon, \tau)$ by $\varepsilon \partial w_i / \partial F(y_i) = \varepsilon \dot{w}_i$. The indirect contribution is therefore $\frac{1}{n} \sum_{\{i: y_i \geq y_j\}} \dot{w}_i \mathbf{x}_i \hat{e}_i$, where $1/n$ arises from the discrete CDF step.

Step 4: Assemble. $\psi_j = A^{-1} \left[\hat{w}_j \mathbf{x}_j \hat{e}_j + \frac{1}{n} \sum_{\{i: y_i \geq y_j\}} \dot{w}_i \mathbf{x}_i \hat{e}_i \right]$. □

B.4 MSE-optimal bandwidth derivation

Bias. $\text{Bias}[\hat{\beta}_j(\tau, h)] \approx \frac{1}{2}\beta_j''(\tau) \int (p - \tau)^2 K\left(\frac{p-\tau}{h}\right) dp = \frac{1}{2}\beta_j''(\tau) \cdot h^2 \mu_2(K) = h^2 \beta_j''(\tau)$.

So $\text{Bias}^2 = h^4 [\beta_j''(\tau)]^2$.

Variance. $\text{Var}[\hat{\beta}_j(\tau, h)] \approx \frac{\sigma_j^2(\tau)}{n} \cdot \frac{R(K)}{h} = \frac{\sigma_j^2(\tau)}{2nh\sqrt{2\pi}}$.

Optimisation. $\text{MSE}(h) = h^4 [\beta_j''(\tau)]^2 + \sigma_j^2(\tau)/(2nh\sqrt{2\pi})$. $\partial \text{MSE} / \partial h = 4h^3 [\beta_j''(\tau)]^2 - \sigma_j^2(\tau)/(2nh^2\sqrt{2\pi}) = 0$. Solving: $h^5 = \sigma_j^2(\tau)/(8n\sqrt{2\pi}[\beta_j''(\tau)]^2)$, giving (10). □

C The gepwreg Stata Command

gepwreg implements the full PWR pipeline. Requires Stata 16+.

C.1 Syntax

`gepwreg depvar [indepvars] [fw/aw/pw] [if] [in] , [options]`

Table 11: Options of `gepwreg`

Option	Default	Description
<code>per(#)</code>	0.5	Target percentile $\tau \in (0, 1)$
<code>cband(#)</code>	0.9	Silverman constant c_0
<code>band(#)</code>	0	Manual bandwidth (0=Silverman)
<code>boot(#)</code>	0	Bootstrap replications
<code>vce(svy)</code>	–	Taylor SE (requires <code>svyset</code>)
<code>rankvar(z)</code>	–	Rank observations by z instead of y
<code>level(#)</code>	95	Confidence level
<code>noconstant</code>	–	Suppress constant

C.2 Examples

Example 1: Basic usage.

```
gepwreg lexp size male urban i.gse [pw=weight], per(0.5)
gepwreg_setable
```

Example 2: Survey design.

```
svyset psu [pw=weight], strata(strata)
gepwreg lexp size male urban i.gse, per(0.5) vce(svy)
```

Example 3: Generalised ranking variable.

```
/* Effects on consumption for households poor in income terms */
gepwreg lexp size male urban i.gse [pw=weight], ///
    per(0.25) rankvar(income)
```

Example 4: Profile $\hat{\beta}(\tau)$.

```
matrix B = J(9,3,.)
forvalues i=1/9 {
    local tau = 'i'/10
    gepwreg lexp size [pw=weight], per('tau') vce(svy)
    matrix B['i',1]='tau'
    matrix B['i',2]=_b[size]
    matrix B['i',3]=_se[size]
}
svmat B, names(col)
gen lo=B2-1.96*B3
gen hi=B2+1.96*B3
twoway (rarea lo hi B1, color(blue%15)) ///
    (line B2 B1, lcolor(blue)) ///
    , xtitle("Percentile {&tau}") ytitle("{&beta}({&tau}) -- size")
```

Example 5: Bandwidth sensitivity.

```
foreach cb in 0.5 0.7 0.9 1.1 1.3 {  
    gepwreg lexp size [pw=weight], per(0.5) cband('cb')  
    di "cband='cb'  h=" %6.4f e(h) "  N_eff=" %5.1f e(N_eff) ///  
    "  beta=" %8.5f _b[size]  
}
```

C.3 Stored results

Table 12: Results stored in `e()` after `gepwreg`

Type	Name	Content
Scalars	<code>e(N)</code>	Observations
	<code>e(tau)</code>	Target percentile
	<code>e(h)</code>	Bandwidth
	<code>e(N_eff)</code>	Kernel effective sample size
	<code>e(do_svy)</code>	1 if <code>vce(svy)</code>
Matrices	<code>e(b)</code>	Coefficients
	<code>e(V)</code>	Main variance (IF or SVY)
	<code>e(V_naive)</code>	Naive WLS variance
	<code>e(V_boot)</code>	Bootstrap variance
	<code>e(V_svy)</code>	Taylor survey variance
Macros	<code>e(cmd)</code>	<code>gepwreg</code>
	<code>e(vce)</code>	SE type
	<code>e(wgt)</code>	Weight variable

C.4 Installation

```
/* Copy .ado, .sthlp, .dlg to Stata adopath */  
which gepwreg    /* verify */  
db    gepwreg    /* dialog */  
help  gepwreg    /* documentation */
```